

Class Note: Introduction to `java.util` Package

Course Title: Advanced Java Concepts

Session: Java Utility APIs

Topic: Introduction to `java.util` Package

Duration: 15 minutes

Learning Objectives

By the end of this lesson, learners should be able to:

1. Understand what the `java.util` package is and why it is important.
2. Identify key classes and interfaces contained in the package.
3. Import and use some basic utility classes in simple programs.

1. What is the `java.util` Package?

The `java.util` package is one of the **core utility packages** in the Java Standard Library. It provides **utility classes and interfaces** that are essential for day-to-day programming tasks.

It contains:

- **Collection Framework classes** – for handling groups of objects such as lists, sets, and maps.
- **Utility classes** – for date, time, random numbers, and more.
- **Legacy classes** – such as `Hashtable`, `Stack`, and `Vector` (still supported for backward compatibility).

 **Think of `java.util` as Java's toolbox** — it gives you pre-built tools for managing data, performing operations, and simplifying complex tasks.



2. Major Components of java.util Package

a. Collections Framework

The Collections Framework is the heart of the `java.util` package. It provides **interfaces** and **classes** for storing and manipulating groups of objects.

Category	Interfaces	Common Implementations	Description
List	List	ArrayList , LinkedList , Vector	Ordered collection, allows duplicates
Set	Set , SortedSet	HashSet , TreeSet , LinkedHashSet	Unordered, no duplicates
Queue	Queue , Deque	LinkedList , PriorityQueue	FIFO (First-In-First-Out) structure
Map	Map , SortedMap	HashMap , TreeMap , LinkedHashMap , Hashtable	Key–value pair collections

b. Utility Classes

Some non-collection utility classes commonly used include:

Class	Description	Example Use
Arrays	Utility methods for working with arrays	Sorting, searching, converting arrays to lists
Collections	Utility methods for working with collections	Sorting, reversing, synchronizing collections
Date / Calendar	For working with dates and time	Getting current date/time
Random	For generating random numbers	Simulating games, lottery, etc.
Scanner	For reading input from user or files	Reading console input
StringTokenizer	For splitting strings into tokens	Parsing CSV or command-line input



3. Importing and Using `java.util` Classes

To use any class from this package, you must **import** it at the top of your program using one of two methods:

Option 1: Import a specific class

```
import java.util.Scanner;
```

Option 2: Import the entire package

```
import java.util.*;
```



4. Practical Examples

Example 1: Using the Scanner class

```
import java.util.Scanner;

public class InputExample {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter your name: ");
        String name = input.nextLine();
        System.out.println("Hello, " + name + "!");
        input.close();
    }
}
```

● Explanation:

The `Scanner` class from `java.util` allows reading user input easily from the console.

Example 2: Using Random class

```
import java.util.Random;

public class RandomExample {
    public static void main(String[] args) {
        Random rand = new Random();
        int number = rand.nextInt(10); // Generates number between 0-9
        System.out.println("Random number: " + number);
    }
}
```

● Explanation:

The `Random` class generates random values for simulations, games, or testing.

Example 3: Using Arrays class

```
import java.util.Arrays;

public class ArraysExample {
    public static void main(String[] args) {
        int[] numbers = {5, 3, 8, 1};
        Arrays.sort(numbers);
        System.out.println("Sorted numbers: " + Arrays.toString(numbers));
    }
}
```

● Explanation:

The `Arrays` class provides static methods like `sort()` and `toString()` to handle arrays efficiently.

5. Key Takeaways

- ✓ `java.util` is a core utility package in Java containing powerful tools for handling data.
- ✓ The **Collections Framework** is the most important part of `java.util`.
- ✓ Utility classes like `Scanner`, `Random`, and `Arrays` simplify common programming tasks.
- ✓ Always **import** the package before using its classes.

6. Quick Quiz (Check Your Understanding)

1. What is the purpose of the `java.util` package?
2. Name any **three** classes found in the package.
3. What is the difference between `List` and `Set` ?
4. Which `java.util` class is used for reading user input?
5. What method in the `Arrays` class can you use to sort elements?

7. Assignment / Practice Task

Create a Java program that:

1. Reads five numbers from the user using `Scanner` .
2. Stores them in an array.
3. Sorts them using the `Arrays.sort()` method.
4. Displays the sorted numbers.